Machine Markets: Proof and Evidence Update

Research completion record

October 9, 2026

This update reports new proof work, primary-literature comparison, inference calculations and displayed-book calibration. It supplements the preserved author/source-recovery release. Full manuscript formalization and real-account calibration remain incomplete. The main manuscript source has been revised at the corresponding literature, inference and financial-calibration passages. The execution runtime is working again on October 9. The strict audit now checks 150 named mathematical declarations. A revised 54-page manuscript PDF has been exported and its edited passages visually inspected; it is a review draft, not a completed submission certificate.

1 Precise contribution

The proposed increment is the joint specification of the observer, execution resource and financing account. Two known latent cohort configurations produce scalar price observations. For deterministic probes, a lifetime absolute-quantity budget has an exact finite-horizon signal optimum. Combining the resolution calculation with a self-financing account gives a distinct capital order. Exact marked-equity disclosure supplies a constructive counterexample to a universal capital barrier.

The relevant observer is an outside analyst receiving noisy aggregate price reports and probe quantities without the financing trader's exact execution/account records. An account owner with exact marks has a different experiment. Quotation residuals mix dynamics, specification error and other noise; they do not identify the theorem's independent measurement-noise variance.

Primary comparison	Established contribution	Proposed distinction
Chatzikiriakos–Iannelli (2024)	Finite-model likelihood and information sample bounds with Gaussian excitation and process noise.	Scalar price reports, latent cohort alternatives, lifetime executed quantity and financial accounts.
Chatzikiriakos et al. (2026)	Fully observed dynamics, average input power, excitation design and active learning.	Fixed lifetime quantity versus power, exact pulse optimum and disclosure/account boundaries.
Alfonsi et al. (2016)	Transient impact, positive-definite kernels and execution-cost structure.	Compose probe execution cost with the particular noisy-price memory-resolution experiment.
Sornette et al. (2023)	Non-normal financial dynamics and Kreiss diagnostics.	Retain explicit monetary output and execution/account geometry.

No first-use claim is made for Gaussian testing, quadratic execution costs, finite-model identifi-

cation, excitation design or non-normality in finance. The comparison supports a precise proposed increment; it does not certify priority or scholarly importance. Unknown-mixture estimation, adaptive-probe optimality and identified public-market learning remain outside the central theorem.

2 New formal proof work

The current strict receipt checks 150 named mathematical declarations and permits only the standard Lean foundations `propext`, `Classical.choice` and `Quot.sound`. Every declaration passes the Lean kernel and dependency audit; no additional project axioms or unfinished proof placeholders are accepted. The earlier 122- and 138-declaration receipts are preserved. The completed additions comprise:

- Thirteen results connect the actual price-first cohort step, reset recursion and finite price record to causal convolution, the physical-pair Gaussian KL quantity bound and first-pulse attainment, coordinate powers, physical impulse equivalence and its peak bound.
- Three results establish counterparty margin at every later mark following a positive pulse, composition with the quadratic quantity cap, and an arbitrarily small margin-compliant pulse whose exact third equity mark distinguishes the known alternatives.
- Twelve results establish positive-step order preservation, the finite dealer cash/equity identity, adverse quadratic-entry and displacement bounds, the necessary and sufficient robust nonnegative quantity allowance, its quadratic-root form and every partial last fill.

The sequence-account result quantifies over every finite nonnegative probe sequence and every final-fill fraction in $[0, 1]$. It uses a nonincreasing kernel with direct response $h_0 = \kappa$ as an explicit precondition. Its account inequality and sharp pulse necessity are proved, rather than assumed. Constructing that nonincreasing kernel from the physical cohort transition and composing the trajectory/account timing remain outstanding. This result does not provide a signed-probe or adaptive-strategy guarantee.

The previously missing all-horizon recursion-to-convolution bridge and all-time pulse-margin composition now pass the strict audit. Remaining full-coverage obligations include Gaussian minimax/testing bounds, induced ℓ_2/Parseval composition, the positive-cohort spectral representation, Kalman conditional-MMSE/filtration, merged realization equivalence, passive-excitation moments/testing, full capacity/Jury results, Kreiss/network proofs and the carried-state window gate. These remain assigned to Codex. A subresult count is not a percentage of manuscript completion.

3 Dependence and equivalence sensitivity

The original 768 reconstructed study units contain 96 sampled dates and four years. Equal-weight annual aggregation gives:

Construction	Mean Q	Annual t_3 95% reference	Sign reference
Legacy	0.0081536	$[0.0064393, 0.0098679]$	0.125
Observed EMA	0.0003807	$[0.0001821, 0.0005793]$	0.125
Raw return	−0.0001625	$[-0.0007691, 0.0004440]$	0.500

The annual t_3 calculation requires approximately independent comparable annual errors. Enumeration of all 16 annual sign patterns instead requires independent symmetric annual errors. Four years do not verify either assumption. The non-rejection under the sign reference is not evidence that learning is absent, and the positive EMA t_3 reference is not a structural learning estimate.

For the one-treated-asset fee comparison, independent four-cell contrast arithmetic recovers the observed-Q effect -0.000105705 and date-cluster uncertainty. Under its t_{23} reference, the minimum symmetric equivalence margin is 0.0008681, or 2.2805 times the magnitude of the pooled observed-Q benchmark. Margins of 10%, 25%, 50%, 100% and 200% of that benchmark fail two one-sided 5% tests; 300% passes. These margins are retrospective sensitivity choices rather than prespecified economic tolerances.

If an independently justified bound Γ covers absolute differential-trend/spillover bias, the interval for a causal contrast must expand by Γ at each endpoint. The panel does not identify Γ. It therefore cannot establish an invariant causal fee effect from a selected non-rejection or a permissive equivalence margin.

4 Observed quote-friction calibration

The authenticated January source contains 3,730,870 snapshots. Selecting the last original snapshot of every observed UTC minute, with validation after selection and no missing-minute interpolation, yields 15,593 valid selected books. Units remain those reported by the provider; exchange account and size-unit authentication are separate.

Observable	10th percentile	Median	90th percentile
Spread, quote units	0.1000	0.1000	0.1000
Spread, basis points	0.04713	0.04813	0.05750
Ten-level bid size	4.4822	19.6920	65.0604
Ten-level ask size	4.3872	19.1710	65.1658
Buy sweep, half displayed size	0.05000	0.05000	0.51409
Sell sweep, half displayed size	0.05000	0.05000	0.50576

Sweep costs are above midpoint per reported size unit. They assume immediate access to all displayed levels; they are hypothetical quote costs rather than realized fills or executable guarantees. Three side-depth fractions (1%, 10% and 50%) are retained for both sides. A separate scalar level-by-level verifier checks all six sweep records per selected minute against the vectorized calculation, together with all pooled and daily summaries. No minute-level independence confidence interval is claimed.

Static displayed-depth slopes are not the dynamic structural impact coefficient κ. Near-zero small-sweep slopes reflect executions remaining at the best displayed quote; they do not establish absent

dynamic impact. The zero-spread scenario omits an observed friction. The measured snapshot frictions now support the revised financial interpretation, while actual execution priority, cancellations, funding, account fees and margin remain separate.

Primitive	What the present evidence supplies	Remaining validation
Quoted spread/depth	Provider-linked quote distributions and hypothetical sweeps.	Real fills, access, units and cancellations.
Dynamic impact/memory	Declared model and reduced-form estimates with restriction failures.	Controlled or otherwise identified structural impact and participant memory.
Observation noise	Assumed independent Gaussian noise in the theorem.	A separately measured reporting-error process.
Fees, funding, margin	Parameters in conditional accounting formulas.	Applicable historical account rules and charges.
Capital/exposure	Exact declared cash and inventory identities.	Actual positions, executions and financing records.

5 Reproducibility and execution status

The completion directory retains the inference and calibration scripts, their JSON outputs, an independent verification script, the primary-research comparison and this source. Raw selected-book arrays remain on T9. The native fallback verifier reads only numeric NPY members, never object arrays or pickle payloads.

The previous run encountered macOS library-load failures. On October 9, Lean 4.34.1, Python 3.11 with NumPy/SciPy, and terminal LaTeX all execute successfully again. The expanded build and strict audit pass. Independent checks pass for 458 rendered manuscript fields, all 11 supplement tables (243 rows and 1,575 cells), source-recovery prose, annual/equivalence calculations, and 93,558 scalar sweep records. The 12 new numeric manuscript/report fields are separately bound to their result files. The revised full manuscript compiles without undefined references or overflowing text. Runtime recovery does not resolve missing account records or replace specialist review. This PDF is a record of completed work and explicit remaining obligations, rather than a certificate of full manuscript verification. No submission, push or reviewer message was sent.